

Nekhil Nagia

Website: nekhilnagia.com

nekhil.nagia@mail.utoronto.ca

GitHub: [nicknagi](#)

EXPERIENCE

Amazon Web Services SDE Intern

May 2020 – August 2020 | Toronto

- Worked on the [RDS Aurora Control Plane](#) team responsible for managing and developing control plane components on all provisioned RDS Aurora clusters
- Improved Aurora QA for [availability](#) by adding over [60 failure injection and additional stress tests](#)
- Successfully integrated work with release pipelines thus ensuring Aurora availability SLAs and prevent regressions during new releases

Square Emerging Machine Learning Developer

Feb 2020 – May 2020 | Toronto

- Worked on the [Risk Data Science](#) team to develop solutions to mitigate fraud in Cash App
- Worked on an end-to-end deep learning pipeline to predict chargebacks using raw event level customer data, used tools such as [TensorFlow](#), [SQL](#), [Apache Airflow](#)
- Built an [AI receptionist](#) system called Deep Doorman David as a [side project](#) that uses a custom [face recognition](#) model to interact with visitors at the office and notifies the appropriate employees on Slack

Dessa Software Engineering Intern

May 2019 – Feb 2020 | Toronto

- Building a product called [Foundations](#) for managing [end-to-end machine learning](#) lifecycle
- Involved in all areas of engineering, from setting up devops infrastructure in Jenkins & AWS to architecture and development using tools such as [Kubernetes](#), [Redis](#) and [Python](#)
- Worked on a feature to allow the user to automatically track [data health](#) and consistency using intelligent data validation techniques and alert them to prevent model degradation

Network Architecture Lab IoT Software Engineer

May 2018 – May 2019 | Toronto

- Worked on the [CVST](#) project (Connected Vehicles & Smart Transportation) at the University of Toronto under [Prof. Leon-Garcia](#)
- Developed high performance web server using [Golang](#) and [Python](#) to implement APIs interfacing and processing [over 100 Gigabytes](#) of data stored in Elasticsearch
- Developed a [computer vision](#) algorithm using [OpenCV](#) in C++ to process data from a real-time [camera stream](#) to detect [free parking spots](#) in a parking lot, achieved [80% accuracy](#) on test data and created interactive visualizations using the ELK stack ([Kibana](#))

PROJECTS

GIS System

2017 | Toronto

- Worked in a team of three to develop a [GIS](#) system using data from [OpenStreetMap](#)
- Used optimized data spatial structures such as R-Trees and custom data buckets to display an interactive map during runtime, and implemented a [search engine](#) for querying information
- Implemented pathfinding algorithms such as [A* search](#) to find optimal routes between two points and solve a variation of the travelling salesman problem

ACTIVITIES

Racquet Sports, Camping & Hiking, Everything Computers, Travelling

EDUCATION

UNIVERSITY OF TORONTO Bachelor of Computer Engineering

Expected Completion April 2021 | Toronto